

Script 2 — Methodology

Slides 8–15 · about 4 minutes

Thank you. I will now explain our method.

Our pipeline has three stages. First, preprocessing, where we collect the evidence for each bug into one case file. Second, classification, where a language model reads that file and gives an ODC label. Third, evaluation.

For each bug, we fetch the bug report, run the failing tests to get the error and stack trace, and take the code around where it failed. We save this in a file called context.json. The classes changed by the fix stay hidden and are used only for evaluation. From this file we make two case files: pre-fix, with every hint about the fix removed, and post-fix, with the same evidence plus the fix.

A condition is a taxonomy combined with a strategy. Our default is the scientific strategy with the seven ODC types plus "Other". We run it with Gemini as our main model and gpt-5-mini as a second model.

Both strategies get the same ODC types, rules and evidence summary. Few-shot also gets five worked examples and answers in one call. Scientific gets five probes and can take up to six turns.

In each turn, the scientific model writes a hypothesis and predicts what evidence must be there. If it has enough evidence, it gives the label. If not, it sends a probe for hidden evidence, reads the result, and starts again. The prediction comes before the evidence, so the model cannot just defend its first guess. We adapted this from AutoSD.

To evaluate, we compare pre-fix and post-fix labels at four levels: the same type, one label listed as the other's alternative, the same ODC family, and Cohen's kappa. First, we count the bugs whose pre-fix and post-fix labels are the same, and divide by all bugs. We call this the observed agreement. Second, we ask how often the two labels would match just by luck. For each type, we multiply how often the pre-fix run

used that type by how often the post-fix run used it, and we add this up over all types. We call this the chance agreement. Kappa is the observed agreement minus the chance agreement, divided by one minus the chance agreement. In other words, it is the agreement we got above luck, out of all the agreement that was possible above luck. Zero means no better than luck, and one means perfect. We need this because most of our bugs are Checking or Algorithm, so two runs can agree a lot just by picking those two types. In our results, we describe it in words, such as agreement well beyond chance.

Why trust the post-fix label? ODC defines the type by the fix, and only the post-fix run sees the fix. Human raters agree much more when they see the change. A buggy version can hold about nine faults, and the fix points to the exact one. And nothing else changes between the two runs.

Finally, our setup. We used 410 bugs from six projects. With Gemini, we ran both strategies before and after the fix, which gives 1,640 runs, plus 820 zero-shot runs as a baseline. With gpt-5-mini, we reran the thirteen bugs we also studied by hand. Now my teammate will present our results.

For Q&A: Cohen's kappa in our pipeline

$$\kappa = (p_o - p_e) / (1 - p_e)$$

p_o , observed agreement = bugs whose pre-fix label equals their post-fix label $\div$ all 410 bugs.

p_e , chance agreement = for each ODC type, (share of pre-fix labels of that type) $\times$ (share of post-fix labels of that type), added over all types.

Meaning: 0 = no better than luck, 1 = perfect agreement. Read on the Landis–Koch scale: slight, fair, moderate, substantial, almost perfect.

Our numbers (scientific strategy):

$$p_o = 288 \div 410 = 0.702$$

$$p_e = \text{Checking} (198/410 \times 192/410 = 0.226) + \text{Algorithm} (175/410 \times 160/410 = 0.167) + \text{Assignment} (28/410 \times 37/410 = 0.006) + \text{the rare types} \approx 0.399$$

$$\kappa = (0.702 - 0.399) / (1 - 0.399) = 0.303 / 0.601 \approx \mathbf{0.505}, \text{ moderate agreement.}$$

Few-shot: $p_o = 0.666$, $p_e = 0.453$, $\kappa \approx 0.389$, fair. Its chance agreement is higher because it labels 265 of 410 bugs as Algorithm, so raw matches alone would flatter it.

One sentence: “The labels match on 70% of bugs where luck alone would give about 40%, so we capture about half of the agreement that luck cannot explain.”